🚀 SE LAB 10-MINUTE PROJECT PROPOSAL

ShareBite BD

Smart Surplus Food Rescue & Safety-Compliant Redistribution Network

📍 Pilot: UIU Campus & Badda | 🎯 UN SDG 2 (Zero Hunger) & 12 | ⚡ 15–30 Min Rescue Velocity

Course: Software Engineering Lab (CSE 4xxx) • Tech: Next.js, Prisma, Leaflet GIS

Team Roster: [Member 1 (Lead), Member 2 (Backend), Member 3 (GIS), Member 4 (QA) | Student IDs: 011 2X1 XXX]

1. Problem Identification: What Problem Are We Solving?

The critical paradox of urban food waste vs. street hunger in Dhaka City



🔴 The Daily Food Dump

Every night, university cafeterias (like UIU), caterers & restaurants throw massive pots of safe, cooked meals into city dumpsters due to zero instant charity links.

⏳ The 4-Hour Biological Clock

Cooked food enters the bacterial danger zone (5°C–60°C). Under BFSA guidelines, food spoils before slow manual phone calls can coordinate a pickup.

🚫 The Missing Logistics Link

Charities lack standby transport at 10 PM. Food exists, hungry people are 2 km away, but zero automated micro-logistics platform connects them in time.

1. Evidence & Data: Backed by Research & Real Impact

Authoritative statistics, biological thresholds, and social impact indicators

10.7M Tons
Food Wasted / Year
UNEP Food Waste Index (BD)

4 Hours
Biological Window
BFSA Danger Zone (5°C-60°C)

3.8M People
Urban Food-Insecure
BBS & World Bank Slum Data

0% Digital
Traceability Today
Zero Audit Trail in Charity



🌍 Tangible Real-World Impact (Key Takeaways):

- **Zero Hunger (SDG 2):** Feeds orphanages, street children & laborers with preserved human dignity.
- **Food Safety Enforcement:** Strict BFSA thermal holding standards (>65°C hot, <4°C cold) prevent foodborne illnesses.
- **Climate Action (SDG 12):** Rescuing food stops rotting in open landfills, eliminating potent urban methane emissions.

2. Solution: The Core Idea & System Approach

A synchronized 3-sided micro-logistics platform engineered for 15–30 minute rescue velocity

☀️ ONE SINGLE LINE (FOR A KID OR PARENT):

"An Uber-like app where university cafeterias post extra hot food, and nearby charities and student volunteers get alerted to pick it up and feed hungry people before it gets cold and goes bad!"

The 3-Sided Hyper-Local Ecosystem:

🍛 1. Food Donors

Cafeterias & Caterers

- 60-Sec listing form
- AI Spoilage & Safety Meter
- Automated meal quantity calculator

🏢 2. Verified NGOs

Charities & Shelters

- Proximity Radar Map (3–5 km)
- 1-Click rescue claim
- Digital QR delivery verification

🚴 3. Student Volunteers

Campus Biker Squad

- Uber-style instant dispatch
- Turn-by-turn route navigation
- Insulated thermal carrier check

2. Main Features: What Will the End System Perform?

Idea-driven key functionalities solving real friction in food rescue



⚡ 1. 60-Sec Listing & AI Spoilage Meter

Live temperature gauge (>65°C safe holding) + countdown clock. Converts weight to child meals (e.g., '40 kg Biryani = 80–100 meals').

📍 2. Proximity Radar Map (3–5 km)

Filters donations strictly within short radius to bypass Dhaka traffic. Batches under 2 hours flash as '🚨 Urgent Rescues'.

🚴 3. Uber-Style Volunteer Dispatch

Crowdsources campus student cyclists and bikers on daily commutes with verified thermal bag checklists.

🔒 4. 4-Digit OTP & QR Code Anti-Fraud

Zero theft: Donors verify volunteer's 4-digit PIN before handover; shelter scans QR token on arrival.

3. Plan of Action: High-Level Technology Overview

Modern open-source stack chosen for velocity, type-safety, and zero hosting cost

💻 Frontend: Next.js 16 + React 19 + Tailwind CSS v4

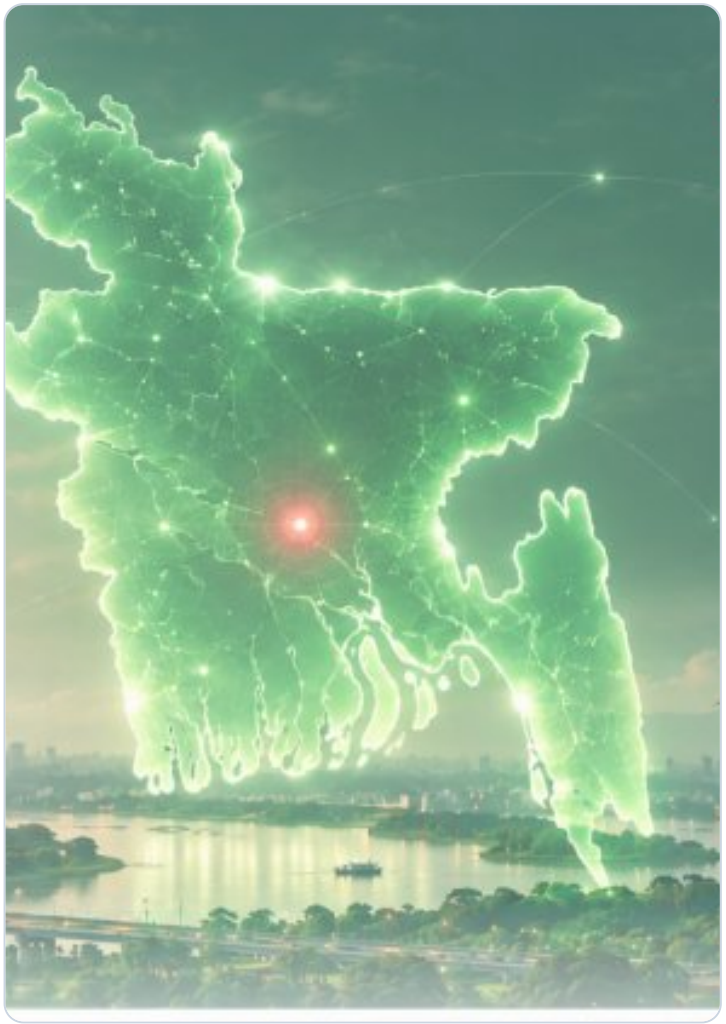
Why: Server-Side Rendering (SSR) delivers sub-100ms load times and full mobile responsiveness without native app installation friction.

🗄️ Backend: Prisma ORM + PostgreSQL + TypeScript / Zod

Why: Strongly-typed relational data models (Users, Donations, Claims, Receipts) with atomic state transitions and zero runtime type errors.

🗺️ Geospatial Maps: Leaflet.js + OpenStreetMap

Why: 100% free, open-source geospatial routing & Haversine distance calculations without requiring expensive Google Maps API billing.



4. Project Timeline: 16-Week Implementation Roadmap

Structured semester milestones leading to a live campus pilot and defense readiness

WEEKS 1 – 4

Sprint 1: Architecture

- Requirements (SRS)
- Prisma DB Models
- Responsive 3-Portal UI
- Role-Based Auth

WEEKS 5 – 8

Sprint 2: Core Engine

- AI Spoilage Countdown
- Leaflet Radar GIS
- 4-Digit OTP Handover
- Claim State Machine

WEEKS 9 – 12

Sprint 3: Campus Pilot

- UIU Cafeteria Live Run
- Badda Local Eateries
- Student Biker Squad
- Mastul NGO Delivery

WEEKS 13 – 16

Sprint 4: Defense

- End-to-End Testing
- CSR Impact Engine
- Security & Load Audit
- Final Defense Deck

🚀 Summary: Zero Waste, Zero Hunger

ShareBite BD addresses food waste with zero hosting cost, an authentic UIU campus pilot, and strict food safety compliance.

Thank You!

Floor Open for Q&A 🤝